

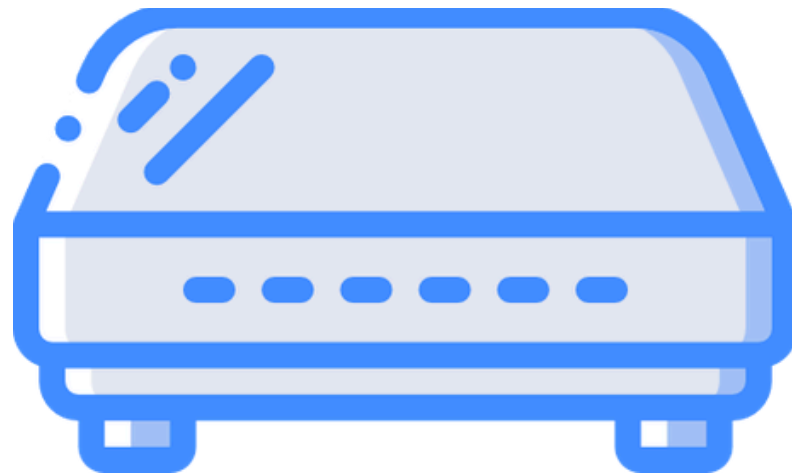
Networking and Communications



Networking Devices & Topologies

Networking Devices

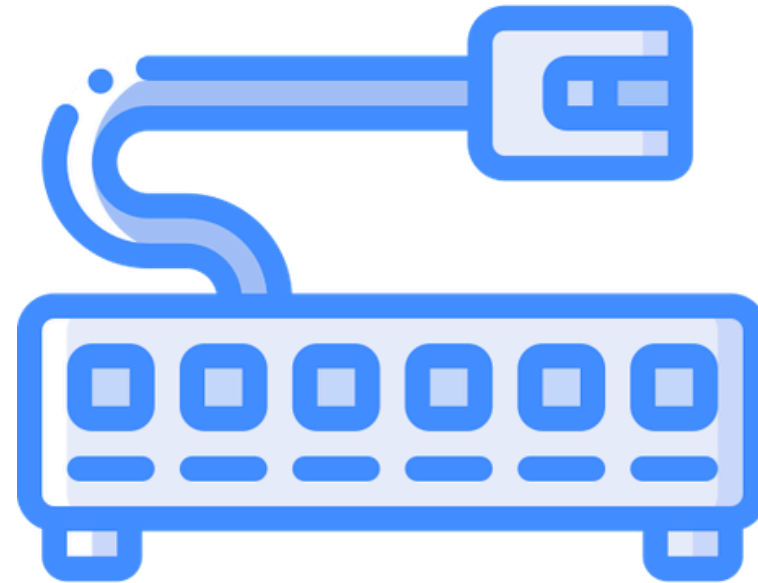
Hubs



Used in Early Networks, also called “Multi-Port Repeater”. Hubs are replaced by Switches because hubs are Dumb Network Devices. Traffic goes in one port and is repeated (broadcasted) out every other port which causes increased network collision errors

Networking Devices

Switches



Connects devices together just like a Hub but the difference is this device is intelligent. Switches memorizes the MAC Address of Each Device connected to it. This fixes the network collision problem commonly seen in Hubs. Switches are standard in today's network infrastructure

Networking Devices

Router



Routes traffic between networks using **IP Addresses instead of MAC addresses. Routers uses Routing Protocols to find the best Way to get a packet of information from One network to another.**

Networking Devices

WAP (Wireless Access Points)



WAPs extend the range of a wireless network by acting as a wireless repeater. This device rebroadcasts radio frequencies from the wireless network it is associated with. Often called as Wi-Fi repeater or Wi-Fi booster, this device expands the coverage of existing internet network. It works by capturing the signal from your main router and then rebroadcasting it to areas of your home or office with a weak or nonexistent signal, known as "dead zones." 🐼

Networking Devices

Range Extender



A wired range extender is a device that extends the reach of your Wi-Fi network using a physical Ethernet cable connection to the main router. This setup is superior to a purely wireless repeater because it avoids the signal degradation and speed loss.

Networking Devices

Firewalls

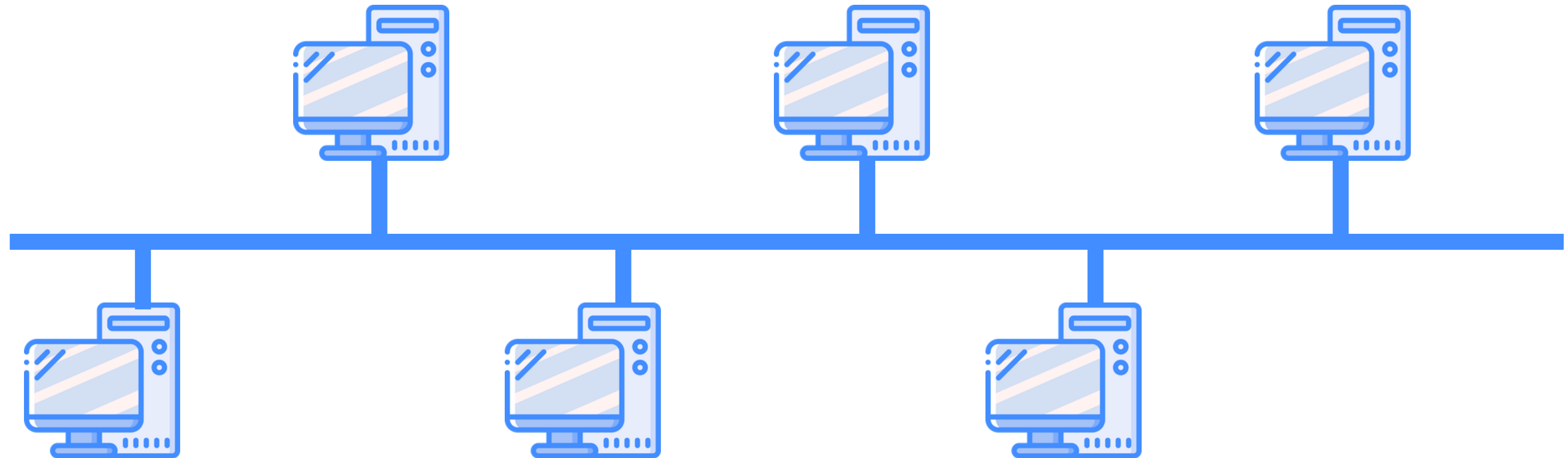


Home segment							
☐	Enabled	Action	Protocol	Source IP	Source port	Destination IP	Destination port
☐	☒	Permit	TCP	Any	Any	Any	80
☐	☒	Permit	TCP	Any	Any	Any	443
☐	☒	Permit	TCP	Any	Any	Any	20
☐	☒	Permit	TCP	Any	Any	Any	21
☐	☒	Permit	TCP	Any	Any	Any	25
☐	☒	Permit	TCP	Any	Any	Any	110
☐	☒	Permit	TCP	Any	Any	Any	143
☐	☒	Permit	TCP	Any	Any	Any	53
☐	☒	Permit	UDP	Any	Any	Any	53
☐	☒	Deny	TCP	Any	Any	Any	Any
☐	☒	Deny	UDP	Any	Any	Any	Any

Firewalls protect your network from malicious activity on the Internet. They do this by preventing unwanted network traffic on different networks from accessing your network using rules.

Wired Topologies

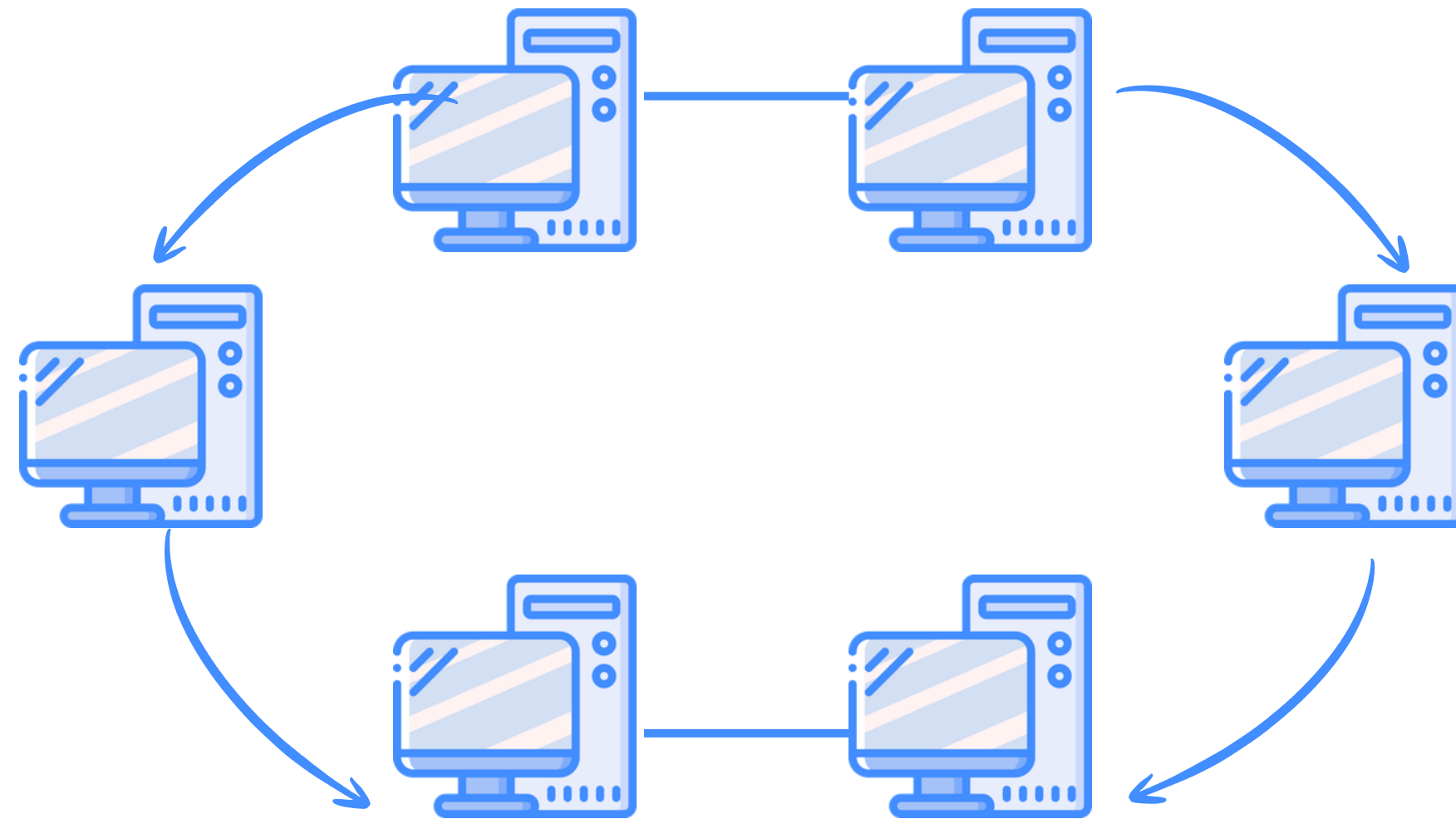
Bus



- **All devices are connected to a single network cable.**
- **Terminators are required at both ends of the cable to prevent signal bounce.**
- **Antiquated technology. Only one end device can be active on the network at a time.**
- **A single break in the cable can take down the entire network.**

Wired Topologies

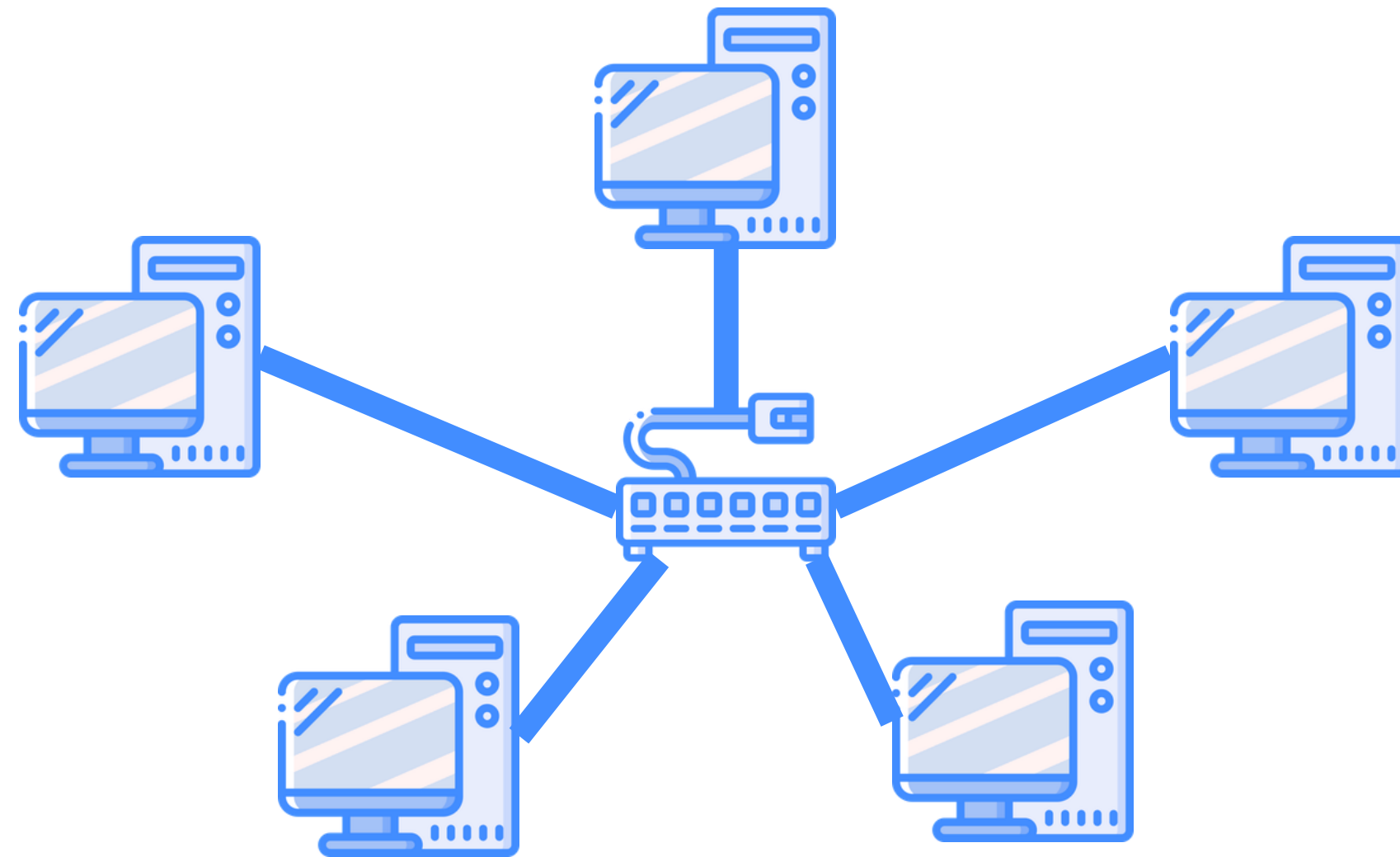
Ring



- **All devices are connected in a circular fashion.**
- **Each computer is connected to two other computers**
- **Data travels from node-to-node either unidirectional or bidirectional.**
- **A single break in the cable can take down the entire network.**

Wired Topologies

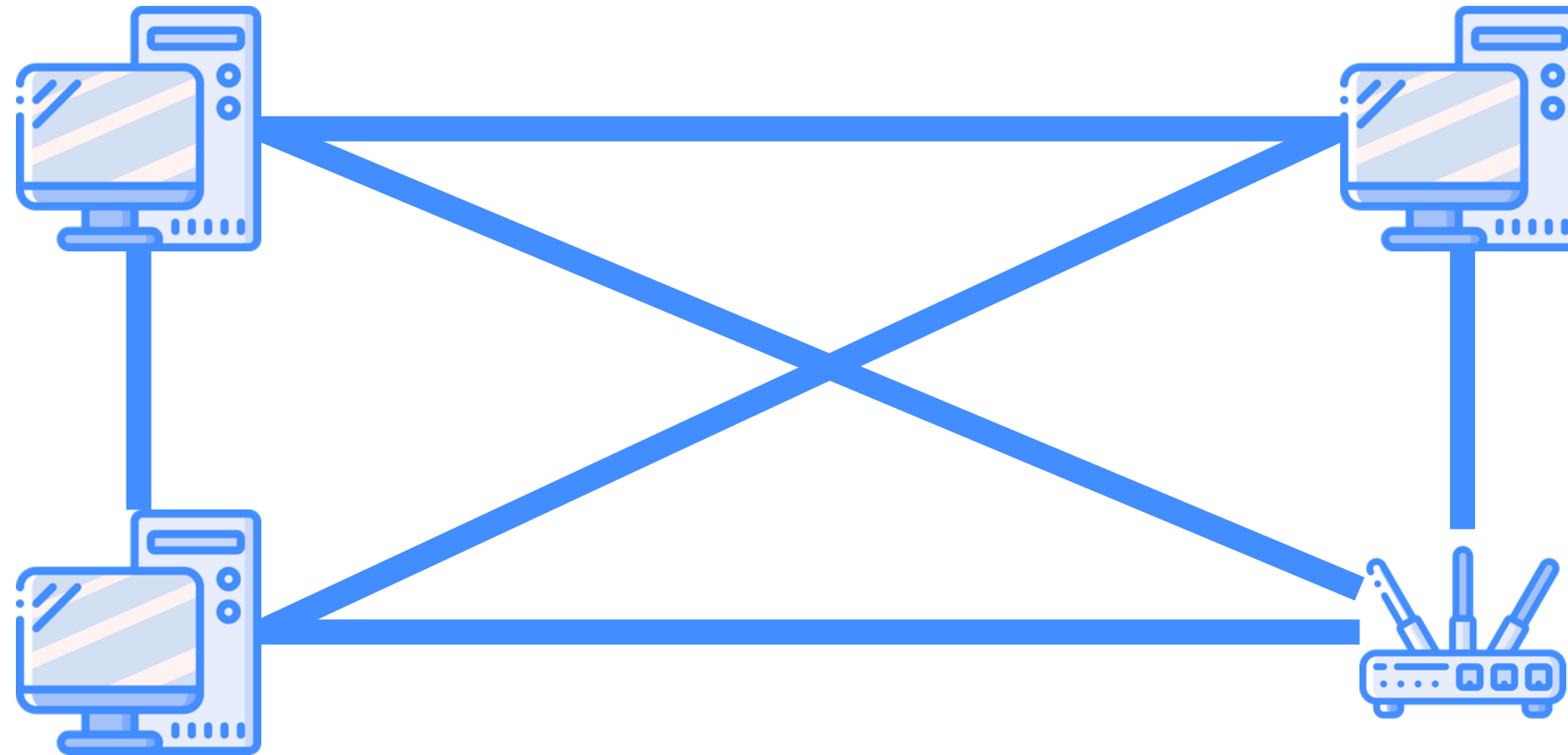
Star



- **Popular topology in today's networks**
- **Used in most large and small networks.**
- **Central device is a single point of failure.**
- **All devices are connected to a central connecting device, which is usually a switch**

Wired Topologies

Mesh



- **Each device is connected to every other device by separate cabling**
- **Highly redundant and fault-tolerance**
- **Expensive to install**
- **Commonly used in Enterprise Networks & WANs**

Wireless Topologies

Ad Hoc



- **Peer-to-peer (P2P) wireless network where no wireless access point (WAP) infrastructure exists.**
- **The devices communicate directly with one another**
- **Personal area networks (PANs) are a common example of Ad hoc wireless networks.**

Wireless Topologies

Infrastructure



- **Wireless network that uses a wireless access point (WAP) as its central connecting device.**
- **Infrastructure wireless networks (WLANs) are commonly used in homes and small offices.**

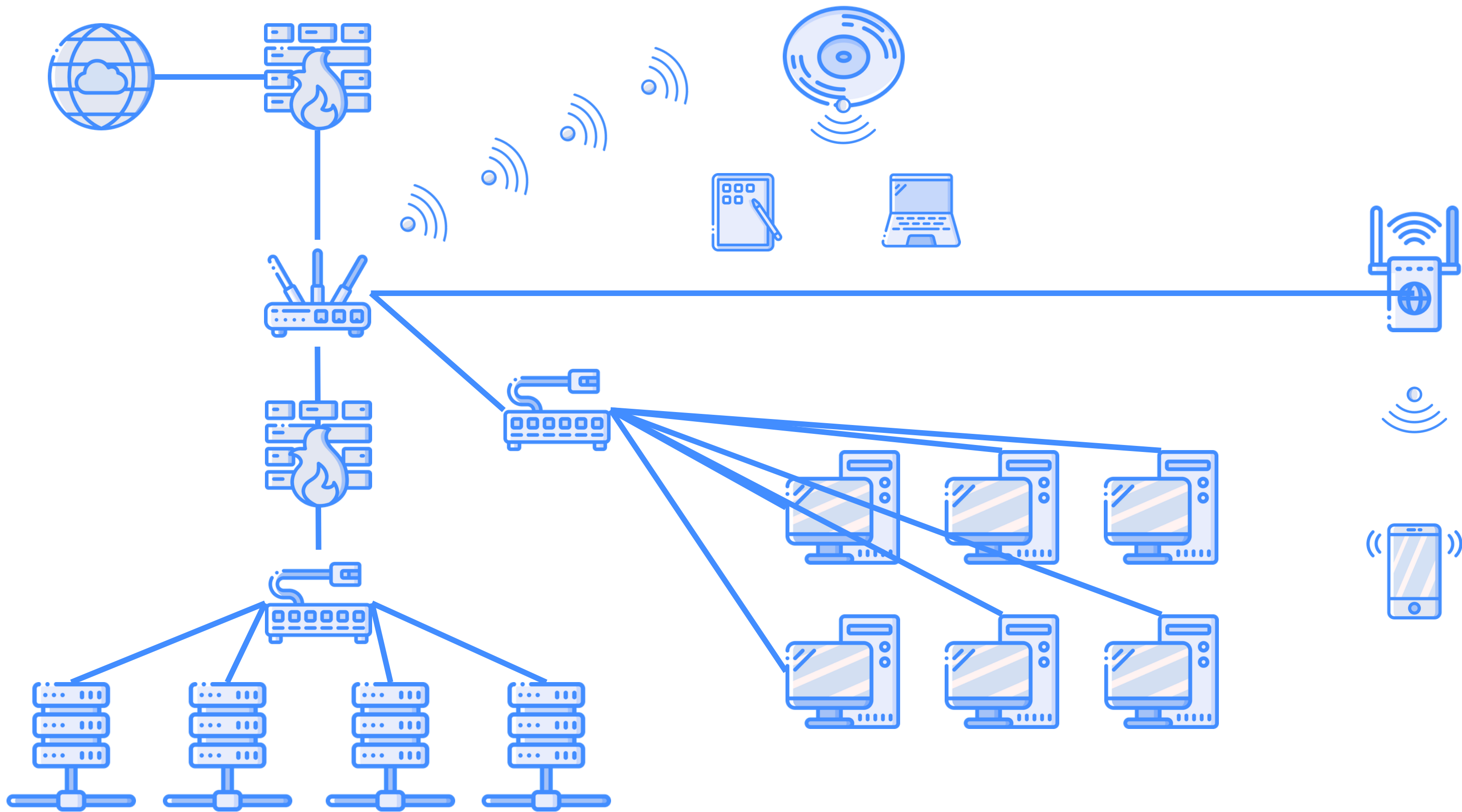
Wireless Topologies

Mesh



- **Just like a wired mesh design, wireless mesh networks utilize several wireless access points (WAP) to create a design that is:**
 - **Scalable**
 - **Self Healing**
 - **Reliable**
- **Common in larger homes and businesses**

Example of network diagram of a small business (Also an Example of Hybrid Topology)



ENCAPSULATION



OSI Model

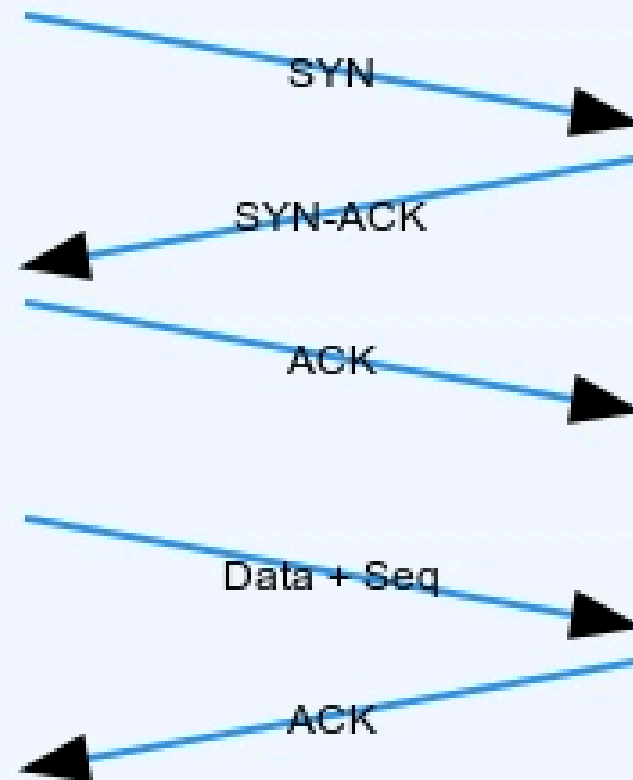
- | | | | |
|----|---------------------|-------------|---|
| 07 | Application | • - - - - - | The closest layer to the user; provides application services. |
| 06 | Presentation | • - - - - - | Encrypts, encodes and compresses usable data. |
| 05 | Session | • - - - - - | Establishes, manages, and terminates sessions between end nodes. |
| 04 | Transport | • - - - - - | Transmits data using transmission protocols including TCP & UDP. |
| 03 | Network | • - - - - - | Assigns global addresses to interfaces and determines the best routes through different networks. |
| 02 | Data link | • - - - - - | Assigns local addresses to interfaces, delivers information locally, MAC method |
| 01 | Physical | • - - - - - | Encodes signals, cabling and connectors, physical specifications. |

TCP vs UDP Protocols

TCP

Client

Server

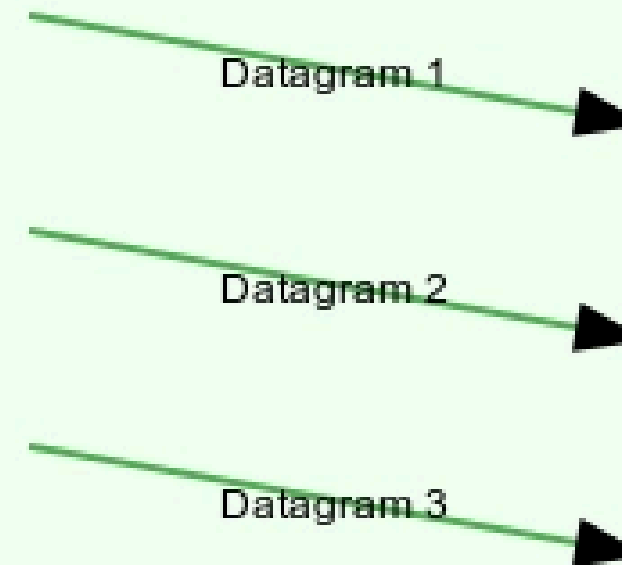


- Connection-oriented
 - Reliable delivery
 - Order guaranteed

UDP

Client

Server



- Connectionless
 - Fast delivery
 - No order guarantee

Trick to quickly understanding
TCP vs. UDP

